A–B. Counting Rules, Arrangements, and Selections

Sum and product rules

Use the **sum rule** for disjoint alternatives: $m + n$ ways to do one of two tasks. Use the **product rule** for a sequence of choices: mn ways when every first choice is followed by n second choices. If alternatives overlap, use inclusion–exclusion:

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

A.1–A.4 worked examples

Disjoint alternatives: five soups or four salads give $5+4 = 9$ lunches. **Successive positions:** two letters followed by three digits give $26 \cdot 26 \cdot 10 \cdot 10 \cdot 10 = 26^2 10^3$ strings when repetition is allowed. **Functions:** choose one of eight images for each of five labeled inputs, giving 8^5. **Overlap:** if $|A| = 19$, $|B| = 20$, and $|A \cap B| = 8$, add both sets and remove the duplicate intersection: $|A \cup B| = 19 + 20 - 8 = 31$.

Generalized pigeonhole principle

Placing N objects into k boxes guarantees some box contains at least

$$\left\lceil \frac{N}{k} \right\rceil$$

objects. Forty keys mapped to twenty hash values guarantee occupancy at least $\lceil 40/20 \rceil = 2$. They do not guarantee 3: two keys per bucket is possible.

Permutations and combinations

When order matters, select and arrange:

$$P(n, r) = n(n-1)\cdots(n-r+1) = \frac{n!}{(n-r)!}.$$

When order does not matter, select:

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}.$$

For repeated object types with multiplicities $n_1, \ldots, n_t$ totaling n, the number of distinct arrangements is $n!/(n_1! \cdots n_t!)$.

B worked examples

A committee is unordered, so three people from ten give $\binom{10}{3}$. Offices are labeled, so president then vice president give $P(10, 2) = 10 \cdot 9$. For `BANANA`, begin with 6! arrangements, then remove the duplicate permutations of the three A's and two N's:

$$\frac{6!}{3!2!}$$

An eight-bit string with exactly three 1s is determined by choosing the three 1-positions; the other five positions are forced, giving $\binom{8}{3}$.

Binomial theorem

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^{n-k} y^k.$$

The coefficient of x^3 in $(1+x)^8$ is $\binom{8}{3}$: choose the three factors that contribute x.

Frequent A–B errors

Do not add sequential choices or multiply overlapping alternatives. Decide whether changing order creates a new outcome before choosing $P(n, r)$ or $\binom{n}{r}$. Divide only for genuinely indistinguishable repetitions.

A–B readiness

Can I name the rule, show an unsimplified expression, explain order, and state what each factor or selected position represents?

C. Constrained Counting

Four reliable strategies

1. **Build directly**: multiply successive legal choices.
2. **Split into disjoint cases**: count each case, then add.
3. **Use a complement**: total minus the forbidden cases.
4. **Choose positions or members**: use binomial coefficients, then fill remaining roles.

State the sample objects and why the cases neither overlap nor leave gaps.

C.1: exact composition

1. Translate "exactly": a five-card hand with exactly two hearts has two hearts and three nonhearts. **2. Choose within each category:**

$$\binom{13}{2}\binom{39}{3}.$$

3. Justify the product: each qualifying hand has one unique pair of hearts and triple of nonhearts, so multiply the two unordered selections.

C.2: count by complement

1. Count all: there are 2^8 eight-bit strings. **2. Identify the complement:** "at least two 1s" fails only for zero or one 1. **3. Subtract the disjoint forbidden cases:**

$$2^8 - \binom{8}{0} - \binom{8}{1}.$$

The complement is shorter because it has only these two cases.

C.3: mixed committee

A five-person group uses disjoint pools of twelve CS majors and eight math majors. Exactly three CS majors therefore requires two math majors. Choose the two unordered subsets in

$$\binom{12}{3}\binom{8}{2}$$

ways. "Exactly" fixes both category counts.

C.4: injective functions

For $f : A \to B$ with $|A| = 5$ and $|B| = 8$, **1. label the five inputs. 2. Enforce injectivity:** after an output is used, it is unavailable. The successive choices are

$$8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 = P(8, 5).$$

3. Check overcounting: input labels already distinguish the positions, so do not divide by 5!.

C.5: codes with a first-position rule

1. Enforce the special position first: a six-letter code beginning with a vowel has five first-character choices. **2. Remove that letter:** without repetition, the remaining ordered choices are $25, 24, 23, 22, 21$:

$$5 \cdot 25 \cdot 24 \cdot 23 \cdot 22 \cdot 21.$$

Each factor now has a visible job: the first enforces the vowel rule, and the others enforce no repetition.

Avoiding overcounting

A counting method is valid only if each desired object is generated exactly once. Ask:

- Can the same outcome arise from two different choices or cases?
- Did I assign an order to an unordered object?
- Did I forget labels that make positions distinct?
- Are my cases exhaustive and disjoint?

Frequent C errors

"At least" usually needs several cases or a complement. "Without replacement" changes later factors. A restriction on the first character must be enforced before counting unrestricted remaining positions. Never use n^r for an injective function unless $r \leq 1$.

C readiness

Can I construct the count two ways, identify possible overlap, justify every factor, and leave a correct unsimplified expression?

D. Probability Models

Sample spaces and events

A **sample space** Ω lists all possible outcomes; an **event** is a subset of Ω. For equally likely outcomes,

$$P(A) = \frac{|A|}{|\Omega|}.$$

Always justify equal likelihood. Useful identities are

$$\begin{aligned} P(A^c) &= 1 - P(A), \\ P(A \cup B) &= P(A) + P(B) - P(A \cap B), \\ P(A \cap B) &= P(A)P(B \mid A). \end{aligned}$$

D.1: coin and die

Model: toss a fair coin and roll a fair die independently. **Sample space:** there are $2 \cdot 6 = 12$ equally likely coin–die pairs. **Favorable count:** heads gives 6, an even die gives 6, and their intersection gives 3. Hence $6 + 6 - 3 = 9$ favorable outcomes and $P = 9/12 = 3/4$. The subtraction removes the duplicate intersection.

D.2: probability identities

If $P(A) = 0.55$, $P(B) = 0.40$, and $P(A \cap B) = 0.25$, then

$$P(A \cup B) = 0.55 + 0.40 - 0.25 = 0.70.$$

Also $B = (A \cap B) \dot{\cup} (A^c \cap B)$, so $P(A^c \cap B) = 0.40 - 0.25 = 0.15$.

D.3: uniform bit strings

Denominator: 2^8 equally likely eight-bit strings. **Numerator:** choose the three positions containing 1s, so $\binom{8}{3}$ strings qualify. Thus

$$P(\text{exactly three 1s}) = \frac{\binom{8}{3}}{2^8}.$$

The numerator and denominator must describe the same sample space.

Conditional probability and independence

For $P(B) > 0$,

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

A and B are independent exactly when $P(A \cap B) = P(A)P(B)$, equivalently $P(A \mid B) = P(A)$ when defined. Independence is a claim to test, not a default assumption.

D.4: without replacement

A bag has four red and six blue tokens. **First draw:** red has probability 4/10. **Update:** after removing a red, nine tokens remain and all six blue tokens remain. Therefore

$$\frac{4}{10} \cdot \frac{6}{9}.$$

The second factor is conditional on the first draw; keeping denominator 10 would describe replacement, not this experiment.

D.5: two different conditionals

If 24 students use Python, 18 use Java, and 10 use both, then

$$P(P \mid J) = \frac{10}{18}, \qquad P(J \mid P) = \frac{10}{24}.$$

The shared numerator is the intersection. The event after the conditioning bar determines the denominator, so reversing the conditional usually changes the value.

Frequent D errors

Do not add probabilities for a non-disjoint "or" without subtracting the overlap. Do not keep a denominator unchanged after sampling without replacement. Do not confuse $P(A \mid B)$ with $P(B \mid A)$ or infer independence from informal wording.

D readiness

Can I name the sample space, choose the correct denominator, use a complement or overlap identity, and distinguish conditional probability from independence?

E. Modeling, Bayes, Expectation, and Interpretation

Total probability and Bayes' rule

If $B_1, \ldots, B_m$ partition the sample space, then

$$P(A) = \sum_i P(A \mid B_i)P(B_i).$$

Bayes' rule reverses the condition:

$$P(B_j \mid A) = \frac{P(A \mid B_j)P(B_j)}{P(A)}.$$

The prior or base rate $P(B_j)$ remains part of the numerator.

E.1: numerical independence test

If $P(A) = 0.6$, $P(B) = 0.5$, and $P(A \cap B) = 0.3$, then $P(A)P(B) = 0.3 = P(A \cap B)$, so the events are independent. Consequently $P(A \mid B) = 0.3/0.5 = 0.6 = P(A)$.

E.2: screening and base rates

Let C be the condition and $+$ a positive test. Then

$$\underbrace{0.95(0.01)}_{\text{true positives}} = 0.0095, \qquad \underbrace{0.08(0.99)}_{\text{false positives}} = 0.0792.$$

Add the two disjoint routes: $P(+) = 0.0095 + 0.0792 = 0.0887$. Then Bayes' rule keeps the true-positive route in the numerator:

$$P(C \mid +) = \frac{0.0095}{0.0887} \approx 0.107.$$

Even a sensitive test can yield many false positives when the condition is rare; the denominator includes both true and false positives.

E.3: factory Bayes model

Factory A supplies 40% with defect rate 3%; B supplies 60% with rate 1%. First compute both joint routes:

$$P(A \cap D) = 0.40(0.03) = 0.012,$$
$$P(B \cap D) = 0.60(0.01) = 0.006.$$

Thus $P(D) = 0.018$ and $P(A \mid D) = 0.012/0.018 = 2/3$.

Expected value

For numerical outcomes x_i with probabilities p_i,

$$E[X] = \sum_i x_i p_i.$$

Use *net* outcomes when a cost is paid. Expected value is a long-run average, not a guaranteed result from one trial.

E.4: expected net value

A fair die pays 10 on a 6, 2 on a 4 or 5, and 0 otherwise. With a 2 cost, first convert payouts to net outcomes: $8, 0, -2$, with probabilities $1/6, 2/6, 3/6$. Therefore

$$E[X] = 8\left(\frac{1}{6}\right) + 0\left(\frac{2}{6}\right) - 2\left(\frac{3}{6}\right) = \frac{1}{3}.$$

The positive sign means an average gain of \$1/3 per play over many plays.

Aggregation reversal

A treatment can have the higher success rate in every subgroup but a lower overall rate when the groups receive treatments in very different proportions. Compute each within-group rate and each pooled rate separately. Then explain which high- or low-success group dominates each pooled denominator; do not say that arithmetic is contradictory.

Twenty-minute model-selection routine

For one problem from each A–E section: name the rule or model first, write an exact unsimplified expression, then calculate or interpret. Check that each denominator matches its condition and that each counted outcome appears once.

Frequent E errors

Do not ignore base rates, reverse a conditional, or assume independence. In expected value, subtract costs from every applicable outcome. In aggregated tables, compare rates rather than raw success counts.

Whole-Core readiness

I can choose among sum, product, cases, complement, permutation, and combination; justify a probability denominator; compute conditional and Bayes probabilities; test independence; calculate net expected value; and explain an aggregation reversal. A live form contains one problem from each A–E section.